UDS
unmanned defense systems



AVENGER 5

Precision Mid-range Strike UAV
TECHNICAL SPECIFICATIONS OVERVIEW

AVENGER 5 is an advanced, cost-efficient loitering munition designed for precision mid-range strike missions, delivering a CEP below 50 cm. Its low visual, acoustic, thermal and radar signatures improve survivability in contested environments, while its steep terminal approach supports effective engagement of targets in complex terrain.

The platform is engineered for electronic-warfare resilience, featuring frequency-hopping communications and dead-reckoning navigation in GNSS-denied conditions. Targets can be detected and engaged using an EO/IR gimbal, visual tracking or geolocation references. Once the target is locked, AVENGER 5 can complete the engagement without continuous radio communication.

Its modular payload architecture supports UDS-provided thermobaric, HE and HEAT warheads, as well as integration of partner-selected payloads. AVENGER 5 is mechanically launched and can be integrated with SitaWare HQ and ATAK for coordinated mission execution.

Warhead availability, integration and use are subject to applicable regulations, export restrictions and end-user requirements. Performance depends on configuration, mission profile and operating conditions.

NSN: registration in progress

AVENGER 5

KEY OPERATIONAL CAPABILITIES

Precision Strike Effect
Carries up to a 5 kg warhead, with HE, HEAT and thermobaric options. EO/IR guidance and visual target tracking enable engagement of stationary and moving targets, with a demonstrated CEP below 0.5 m against stationary targets.

Extended Strike Reach
Up to 90 minutes of endurance provides the reach and time required to locate, confirm and engage targets beyond the immediate frontline.

Autonomous Terminal Engagement
After target lock, AVENGER 5 can complete the terminal attack without continuous radio communication, increasing resilience against jamming and link disruption.

Affordable Precision at Scale
A modular, production-ready strike platform combining interchangeable warheads, mobile launch capability and shared SwarmC2 mission software for scalable operational deployment.



*Depending on the mission's CONOPs (Concept of Operations), geolocation and weather conditions

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PERFORMANCE	
Endurance*	90 min
Cruise speed	31 m/s
Max horizontal speed	42 m/s
Stall speed	24 m/s
Max dive speed	80 m/s
Wind tolerance ground	12 m/s
Wind tolerance cruise	28 m/s
Service ceiling	5000 m
Operating temperature range	-20 to +50°C

STRUCTURAL DETAILS	
Wingspan	1.8 m
Overall length	1.6 m
MTOW	17 kg
Color	NAVY Grey / Camouflage

COMMUNICATION	
Main frequency (different uplink and downlink frequencies)	4.4-5.0 GHz
Radio output power	2 W
Encryption	AES-256
RLOS* / w relay	60 km / 90km
Interference avoidance	Yes

FLIGHT CONTROL	
Autopilot	Fully autonomous waypoint navigation
Safety	Programmable fail safes GNSS denied position correction
Failsafe routes	Dead reckoning, continuous health monitoring

ANTENNA	
Type	Directional and omnidirectional
Functionality	Rotator autonomously tracking UAV

OPERATIONAL MODES	
Deployment	Pneumatic launch system

INSTRUMENTATION	
GNSS	GPS/GLONASS/GALILEO
Inertial Navigation System (INS)	Accelerometers Gyroscope Magnetometer Barometer Heated Pitot tube
Propulsion	Fully electric
Navigation, surveillance and targetting gimbal	Zoom - X40 (X20 + X2 digital) Day camera - 1280x720 Thermal camera - 1280×720 2-axis stabilized Pitch: -45° to +135° Yaw/Roll: -165° to +165°

MISSION CONTROL	
Type	Rugged tablet
OS	Windows 11
Software	UDS SwarmC2

GROUND STATION CONTROL	
Functionality	Mission planning Automatic object tracking and geolocation Integrated video display Direct control of UAV Integration with Battle Management Systems Connect with 200m fiber optics spool

PAYLOAD	
Warhead	5 kg. Thermobaric High explosive (HE) High explosive anti-tank (HEAT)

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